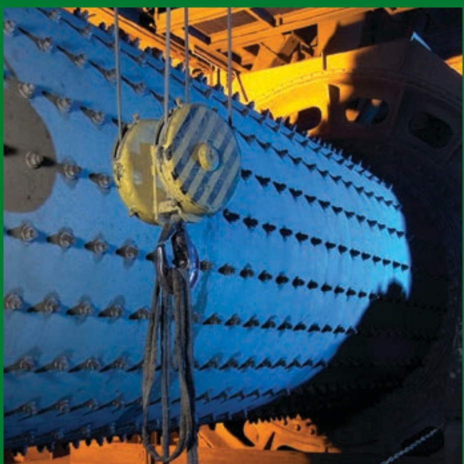
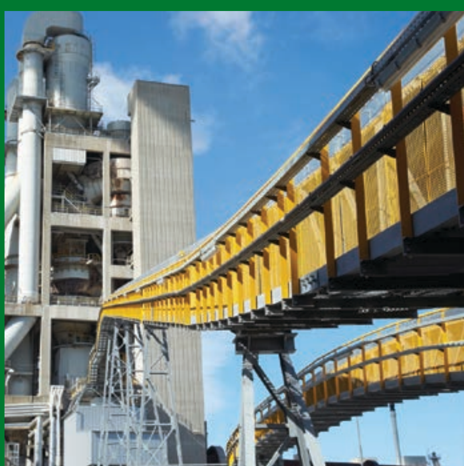
Cement Plant Environmental Handbook

THIRD EDITION








7.7 Synthetic SCMs: a new paradigm of SCM production

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To help avoid the 2˚C rise in global temperature, the cement industry must reduce CO_2 emissions by 24 per cent from the current level by 2050, taking into consideration the projected increase in cement production of 12-23 per cent during this period. Among the methods being explored to lower CO_2 emissions by the cement industry is reduction of the clinker factor using supplementary cementitious materials (SCMs). Solidia Technologies® is leading the way with the development of a synthetic SCM that can significantly reduce the CO_2 footprint of concrete, without compromising on strength.

For some time, cement and concrete industry leaders have sought technologies and practices designed to reduce the CO_2 footprint. Increasingly, concrete companies are joining the quest. Both precast and ready-mix manufacturers are reaching out to industry innovators for solutions that will help them be successful as sustainability up and down the value chain is considered a key element of risk management. Current and developing approaches to achieve this goal include reductions in the clinker factor using supplementary cementitious materials (SCMs) and other innovative new technologies. The cement industry must invest and innovate not only in CO_2 emissions reduction strategies, but also in technologies for direct carbon capture, utilisation and storage (CCUS) in cement and concrete production.

Solidia Technologies® is at the forefront of developing technologies that both reduce CO_2 emissions during cement production and utilise CO_2 during the curing process. Solidia's latest focus is on an alternative SCM developed by carbonating Solidia Cement™. Use of this SCM has the potential to significantly reduce the CO_2 footprint of concrete.

Understanding SCMs

SCMs are added to concrete mixtures for a wide variety of reasons, including improving rheology, pumpability, finishability, durability, and alkali-silica reaction (ASR). SCMs improve the overall hardened properties of concrete through hydraulic or pozzolanic reaction, or both. They are added to the concrete as a partial replacement of ordinary Portland cement (OPC) or directly used in blended cements. SCMs are typically industrial byproducts and their use contributes to environmental and energy conservation goals. The most commonly used SCMs are:

- fly ash, a byproduct of pulverised coal-fired power plants
- ground granulated blastfurnace slag (GGBS), a byproduct of the iron industry
- silica fume from the silicon or ferrosilicon industry.

In addition, small quantities of natural pozzolans are also used, such as calcined clays (metakaolin), volcanic ash, pumice, etc.

The production of Portland cement is a very energy-intensive process and a major contributor to greenhouse gas emissions. World cement production reached 4.1Gt in 2019[1] and is estimated to contribute about eight per cent of total anthropogenic CO_2 emissions.[2,3] During the 21st Conference of the Parties (COP-21) within the United Nations Framework Convention on Climate Change (UNFCC) 2015 in Paris, the member nations adopted by consensus to keep the global temperature rise within 2˚C of the pre-industrial era by the end of the 21st century by reducing the greenhouse gas emissions (The Paris Agreement).[4] To meet this 2 Degree Celsius Scenario (2DS), in its publication, 'Energy Technology Perspectives 2020', the International Energy Agency (IEA) identified and projected pathways towards net zero emissions from 2019 until 2070 for the cement and concrete industries.[5] Reducing the clinker-to-cement ratio, or clinker factor (CF), is identified as a key lever.

The current estimated clinker-to-cement ratio of OPC-based binder in concrete is 70 per cent. The goal is to reduce this ratio to 60 per cent by 2050 to meet the 2DS goals. Beyond 2050, a further reduction in the CF is anticipated, but at a slower pace. SCMs are predominantly used to reduce the CF and are primarily amorphous silicate, aluminosilicate or calcium aluminosilicate powders used as partial replacements of clinker in cements or as partial replacements of OPC in concrete formulations.

The most widely-used SCMs, such as fly ash and GGBS, are industrial byproducts so their consistency in quality is not guaranteed. In addition, supplies of both fly ash and GGBS are diminishing as coal-fired power plants are closing and steel production is modernising. There is also a short supply of SCMs in some markets and the prices have risen. Therefore, there is a need for a synthetic product with consistent quality and comparable performance.

In general, SCMs have a very low CO_2 footprint compared to OPC, as the CO_2 footprint associated with industrial byproducts are allocated to the primary industry. Typically, natural pozzolans have low CO_2 footprints as there are no decarbonation processes involved. Most natural pozzolans only require hydroxylation during the calcination process, which is a relatively low-energy expenditure. However, supply of natural pozzolans is localised and limited. As a direct extension of its CO_2-cured precast

Figure 7.7.1: a sequence of Solidia SCM production – Solidia clinker (left), Solidia Cement (centre), Solidia SCM (right)

Source: Solidia

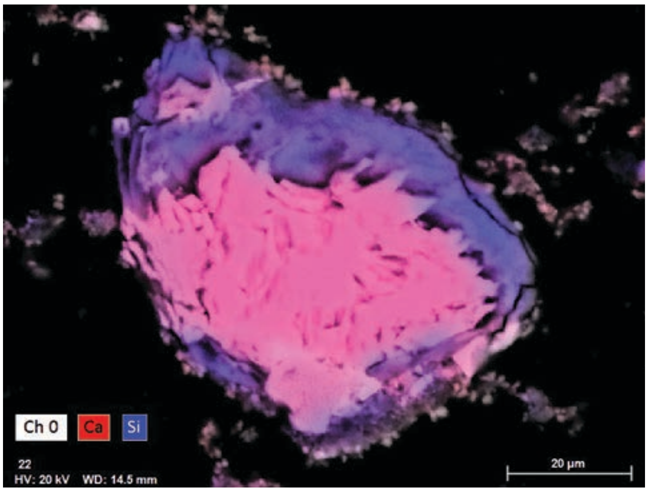
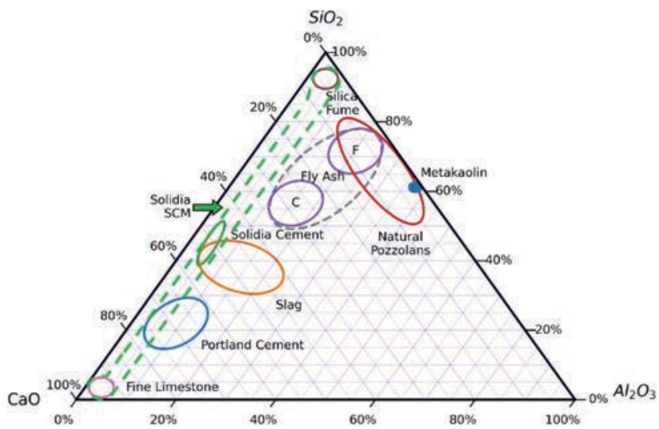


Figure 7.7.2: EDS map of SEM-BSE image showing formation of calcite on the surface as well as away from cement particles and silica rim around the cement particle

Figure 7.7.3: composition of various SCMs in the CaO-SiO_2-Al_2O_3 system



Source: Solidia

concrete application, Solidia Technologies has developed a new technology to produce a synthetic, low-CO_2 SCM that can serve as a substitute for traditional SCMs.

Production of Solidia SCM

Solidia SCM™ is created by directly reacting Solidia Cement™ with CO_2 in a wet (slurry) or semi-wet condition. This carbonation can be done at the cement plant utilising the flue gas, creating an opportunity for direct utilisation of CO_2. In an initial study, industrial-grade CO_2 was used in experiments to carbonate Solidia Cement for the production of Solida SCM in a slurry form. A sequence of Solidia SCM production is depicted in Figure 7.7.1.

The carbonation products of Solidia Cement are described below in Equations 1 and 2. The main carbonation products are calcite and amorphous silica.

$$CaSiO_3 + CO_2 \xrightarrow{(H_2O)} CaCO_3 + SiO_2 \qquad \text{[Eq 1]}$$

$$Ca_3Si_2O7 + 3CO_2 \xrightarrow{(H_2O)} 3CaCO_3 + 2SiO_2 \qquad \text{[Eq 2]}$$

An energy dispersive spectroscopic (EDS) map of scanning electron micrograph (SEM) in the backscattered electron (BSE) mode is shown in Figure 7.7.2. This sample was prepared by mounting in epoxy and polishing. The images show formation of calcite crystals on the surface of the cement particles and silica rim around the core cement particle.

The bulk chemistry of Solidia SCM with respect to other SCMs used in concrete is shown in the CaO-SiO_2-Al_2O_3 ternary diagram in Figure 7.7.3. Although the bulk chemical composition of Solidia SCM is similar to Solidia Cement, it is similar in functionality to slag, silica fume and fine limestone (shown in the ternary diagram with a dotted line connecting Solidia Cement with silica fume and limestone).

The amorphous silica created during this carbonation process is pozzolanic in nature. That is, it reacts with calcium hydroxide produced during the hydration of OPC, creating additional C-S-H gel in the system. The fine calcium carbonate present in this material reacts with OPC to produce a minor amount of monocarbonate phase, as seen in Equations 3 and 4:

$$SiO_2 + Ca(OH)_2 + H_2O \rightarrow CaO\text{-}SiO_2\text{-}H_2O \text{ (C-S-H gel)} \qquad \text{[Eq 3]}$$

$$3CaO.Al_2O_3 + CaCO_3 + H_2O \rightarrow 3CaO.Al_2O_3.CaCO_3.11H_2O \qquad \text{[Eq 4]}$$

The formation of additional C-S-H gel reduces the overall amount of calcium hydroxide in the cement paste. This process helps to improve the strength and durability of the concrete over time.

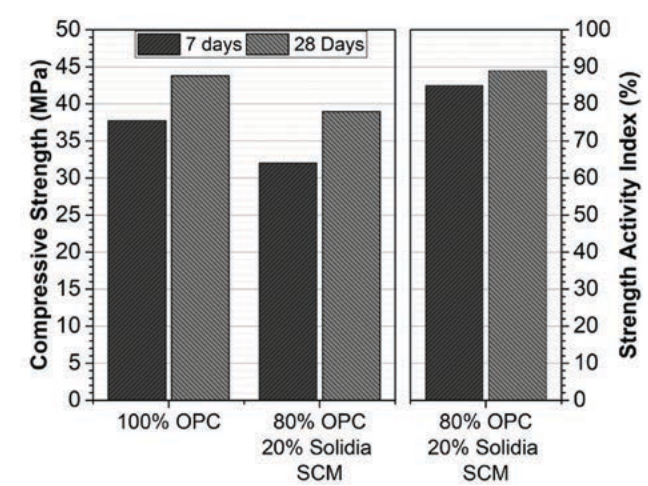
Performance of Solidia SCM

To evaluate the performance of Solidia SCM, the strength activity index (SAI) following ASTM C618 at 20 per cent replacement was carried out. The SAI results are provided in Figure 7.7.4. SAI is an indirect measure of pozzolanic activity of SCMs. The minimum compliance requirement of SAI is 75 per cent at 7 days or 28 days. Solidia SCM meets this SAI requirement at both ages.

Potential reduction in CO_2 emissions

It has been demonstrated that production

Figure 7.7.4: compressive strength and SAI of Solidia SCM at 20 per cent replacement



Source: Solidia

Figure 7.7.5: a demo pour of concrete with 20 per cent Solidia SCM replacement for OPC

of Solidia Cement reduces the CO_2 emissions at the kiln by 30 per cent compared to OPC production.[6] The subsequent carbonation of 1t of Solidia Cement is expected to utilise and store up to 240kg of CO_2 gas, thereby creating 1.24t of a solid, carbonated Solidia SCM product. The creation of Solidia SCM via these two steps represents an opportunity to significantly reduce the amount of OPC used in concrete and, hence, the CO_2 footprint associated with OPC use. Equally important is the potential to make Solidia SCM in cement plants everywhere around the world. This capability will compensate for both current and future shortages in fly ash and slag supplies. Current tests are exploring the range of CO_2 savings as well as the performance of Solidia SCMs. The results of these tests will be reported in the future. A demo pour of concrete with Solidia SCM at 20 per cent replacement of OPC is shown in Figure 7.7.5.

Conclusion

Cement and concrete manufacturers are jumping onto the sustainability train, further driving innovation. The introduction of a synthetic SCM can help them remain competitive during the worldwide push towards reducing the environmental footprint of heavy industry and mitigate ongoing SCM supply interruptions and price fluctuations. Produced through the carbonation process of Solidia Cement, including using flue gas from a cement plant, Solidia SCM is a synthetic material that can be produced in any cement plant with consistent quality. The production of this SCM and application in concrete has the potential to reduce the CO_2 footprint significantly compared to OPC concrete. Preliminary results of tests underway show that Solidia SCM substituted OPC mortar meets the strength activity index requirement.

Acknowledgement
Part of this work is supported by the US Department of Energy Award Number: DE-EE0009417.

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"Cement and concrete manufacturers are jumping onto the sustainability train, further driving innovation. The introduction of a synthetic SCM can help them remain competitive during the worldwide push towards reducing the environmental footprint of heavy industry and mitigate ongoing SCM supply interruptions and price fluctuations."

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